

## RESEARCH REPORT

# Cranial chaos: An interactive and practical overview of cranial nerve functions

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## Abstract

The pressures from a rapidly evolving technological era have brought about a widespread attention crisis in academia; hence, more non-traditional approaches like game-based pedagogies in medical education have become imperative to increase students' engagement. The Cranial chaos card game was part of an integrated laboratory session on Neuroanatomy structure for first-year medical students, introduced in 2022 and repeated in 2023 to review the sensory, motor, and autonomic functions of the cranial nerves. The students completed an anonymous electronic end-of-course evaluation to assess the ease of use, interactivity, and effectiveness of the cranial nerve game in enhancing their understanding of cranial nerve functions. A total of 202 first-year medical students participated in this study. 90.2% of the 184 respondents agreed or strongly agreed that the cranial nerve card game was easy to use, 91.8% agreed or strongly agreed that the cranial nerve card game was interactive, and 78.3% agreed or strongly agreed that the cranial nerve card game facilitated a more practical understanding of cranial nerve functions. Cranial chaos is an innovative, simple, and inexpensive teaching tool for reviewing cranial nerve functions using real-life scenarios.

## KEYWORDS

anatomy, cranial nerves, game-based learning, medical education, serious game

## INTRODUCTION

Traditionally, anatomy had been taught through didactic lectures and laboratory dissections or demonstrations,<sup>1</sup> including cranial nerves which are the cornerstone of neuroanatomy for undergraduate medical students. A study by Lone et al.<sup>2</sup> showed that a group of dental students perceived cranial nerves to be a difficult topic due to the number of cranial nerves that need to be studied, the convoluted course they follow in the cranium, the various fiber types they carry and their corresponding functions. A survey of students across a variety of other health professions substantiated this, as they also found neuroanatomy to be particularly complex.<sup>3</sup> Moreover, poor understanding of the subject has been linked to the difficulty

in its translation into clinical situations.<sup>3,4</sup> With the advent of the integrated medical curriculum, the last few decades have witnessed more advances in the development of case-based, active learning methods for anatomy content delivery.<sup>5</sup> Some of these techniques consist of visual supplements such as medical imaging, virtual or augmented reality software, videos, and simulation.<sup>6</sup>

Moreover, the pressures from a rapidly evolving technological era have brought about widespread attention crisis in academia; hence, more non-traditional approaches to education have become imperative to increase students' engagement.<sup>7</sup> Foremost among these trending teaching approaches is game-based learning.<sup>7</sup> The scope of game-based learning includes serious games, which refer to games that are specifically designed to align with

predetermined learning objectives.<sup>8</sup> Serious games have become increasingly popular in medical education as a fun approach to active learning, bridging the gap between theoretical understanding and an immersive learning environment based on real-life scenarios.<sup>9,10</sup> This has been widely reported to increase class participation, promote collaboration, communication, and effective learning outcomes.<sup>11–14</sup>

Some studies have described various low- and high-fidelity games used as teaching aids to enhance the learning of cranial nerves, including the Cranial Nerve Skywalk<sup>15</sup> and a Charade-type cranial nerve game.<sup>16</sup> The Cranial Nerve Skywalk is a virtual, collaborative learning platform where users exist as 3D avatars and can visualize complex cranial nerve structures, appealing to those more adept in technology.<sup>15</sup> Although this high-fidelity game reported favorable student usability and opinions, it may not be accessible to students and institutions with limited resources or access to technology. Conversely, the Charade-type cranial nerve game is a low-fidelity, analog card game where the players guess the cranial nerves described by teammates.<sup>16</sup> While this game could be easily replicated as it only involves laminated cards, the lack of complexity limits its use to review of cranial nerve function.

In this paper, we describe cranial chaos—a fun, engaging, and easily accessible cranial nerve card game as a learning tool for the review of cranial nerve function. Unlike the charade-type card game where players only guess which cranial nerve is being acted out, cranial chaos simulates humorous real-life scenarios with unique player-driven justifications which promote problem-solving and critical thinking skills. It also introduces randomized, incremental challenges to explore cranial nerve foramina and pathways—an important element in game design. The theoretical framework is based on the self-determination theory, a theory of motivation based on the learners' need for competence, autonomy, and relatedness.<sup>17</sup>

## METHOD

The activity described here was part of an integrated laboratory session on Neuroanatomy structure which occurred during “Challenges, Privileges, and Responsibilities (CPR),” our integrated introductory course touching on all the major bodily systems during the First 100 Weeks. The Neuroanatomy structure session was scheduled for the sixth week of the CPR course, which occurs in the fall of the first year of medical school. CPR is the first course that the new medical students experience. It is a survey course where each week the students learn about a different body system. The design of the complete laboratory session consisted of four complementary stations, including (1) cranial nerves and skull base, (2) brain, (3) spine and spinal cord, and (4) neuroimaging. Students rotated in small groups of seven to eight through each station every 30min, adding up to a total of 2h. Students will revisit the cranial nerve during a neuroscience course in their second year.

## Pre-work

Laboratory guides containing the goals, learning objectives, and required resources for the station-based activities were provided to the students a week before the laboratory session ([Appendix 1](#)). Students were required to prepare for the session by employing the suggested text and multimedia resources to study for each station based on the learning objectives provided.

## Cranial chaos

The “cranial nerves and skull base” station was designed as a highly interactive 30-min session aimed at correlating the foramina in the basocranium with the cranial nerve that passes through them, explaining their sensory, motor and autonomic functions. To review the sensory, motor and autonomic functions of the cranial nerves, the card game was introduced in 2022 and repeated in 2023. This activity was allotted 15 min and required one faculty facilitator. The rules of cranial chaos were described by the faculty facilitator prior to gameplay and remained displayed on the projector throughout the duration of the game.

Contained in each set of cranial chaos are two decks of cards needed to play the game. The Nerve Deck—Deck A ([Appendix 2](#)), which is made of 24 cards; two copies of each of the cranial nerves, one copy is written as the full nerve name, the other 12 the roman numeral set. These cards will be the Draw Pile later when discussing game play and will be the cards making up the students' hands of three cards. The Axon Deck—Deck B will have different scenarios or actions written on them; these are the Prompts for the students to answer, example: “Eating the questionable leftovers in the fridge” ([Appendix 3](#)). Also mixed in the Axon deck are six Bonus Cards that can be added to the rounds to require extra information in the students' answers and therefore are worth extra points ([Appendix 4](#)).

## Gameplay

Decks A and B must be separated, shuffled, and placed face down on the table. Two groups made up of three to five students each draw hands of three cards from the Nerve Deck. Whichever player most recently stepped foot in a hospital would serve as the Judge of the first round. The Judge will turn over the top card of the Axon Deck and read the prompt aloud for the remaining players. Each group will then choose a card from their hand and place it face down in front of them. Once all teams have chosen their card, starting from the left of the Judge they will each reveal and defend their choice of cranial nerve for the prompt. Due to only two copies of each nerve being in the Nerve Deck, and randomization of shuffling, the students may not have the most objectively useful cranial nerve to play in that round. This is to encourage creative defenses and reinforce their knowledge of the cranial nerves' functions, including

correlating with bony landmarks and pathways. After listening to all the defenses, the judge chooses the winner. While creative answers are allowed, all responses should be factual. To finish the round, all used nerve cards are shuffled and returned to the bottom of the Nerve Deck; the winner of the round holds onto the Axon card from the round to keep track of their points. The turn of play now moves to the left and the next player becomes the Judge of the next round. The game continues until either all Axon cards are won by players, or if used in a classroom setting, a certain number of rounds or time limit can be assigned.

There are three types of optional Bonus Cards in the Axon Deck, which can be added to the game to make it more challenging. They are brought into play when they are turned over from the top of the Axon Deck, in which case a second card is turned over to receive the prompt for the round. Synaptic Snag is a Bonus Card worth one extra point. When this card is turned over, students must include in their defense where along the pathway that nerve synapses. This detail of course must be correct, regardless of how they twist the prompt to sway the Judge to their cranial nerve pick. The same is true of the Foramina Fumble card, where students must include which foramina their chosen nerve exits the basocranium for an extra point. The final Bonus Card—Pathway Peril is worth two points due to its difficulty. When this card is drawn, students must include the entire pathway of the signal they are describing. A detailed illustration can be found in [Appendix 5](#).

## Evaluation

At the end of the course, the students were sent an anonymous electronic evaluation survey. This consisted of three questions assessing ease of use, interactivity, and effectiveness of the cranial nerve game in enhancing their understanding of cranial nerve functions. There was an open-ended section where the students were asked to comment on the aspects of the activity that worked well, and what could be improved ([Appendix 6](#)).

## Statistical analysis

Descriptive statistics are presented as the number (and percent) of responses (SPSS Inc., Chicago, Illinois, USA, Version 28.0). We

conducted a conventional content analysis of the open-ended comments from the survey<sup>18</sup> and used the self-determination theory as the conceptual framework.<sup>19</sup> Self-determination theory has been used in literature as a framework for utilizing games in education,<sup>20</sup> with three main categories: autonomy (making choices and having fun), relatedness (finding connections with others and to real-world examples), and competence (learning the material). One author (D.O.) coded free-text comments and arranged responses into meaningful categories, typically the length of a word or complete sentence (units of analysis). Next, a second author (V.R.) reviewed the categories. The two authors then met to discuss and resolve discrepancies.

## RESULTS

### Student's perception of the cranial nerve card game

The cranial nerve card game was used as an interactive activity in our cranial nerves and skull base session in 2022 and 2023; hence, we combined the evaluations. We received a combined total of 184 out of 202 responses to our session survey (91.1%). 166 (90.2%) of the respondents agreed or strongly agreed that the cranial nerve card game was easy to use, 169 (91.8%) agreed or strongly agreed that the cranial nerve card game was interactive, and 144 (78.3%) agreed or strongly agreed that the cranial nerve card game facilitated a more practical understanding of cranial nerve functions ([Table 1](#)).

The open-ended part of the survey received 103 comments which were coded and assigned into three categories based on the SDT model: autonomy, relatedness, and competence. Representative comments associated with each theme are presented in [Table 2](#). Students described how the game contributed to their feeling of competence, described as their ability to learn the material ( $N=42$ , 41%), and that it was fun to learn using the game (autonomy,  $N=41$ , 40%). Students described relatedness in two ways: the benefit of connecting the content to real-life examples in unconventional ways ( $N=12$ , 12%) and feeling connected to their classmates ( $N=4$ , 4%). Students also commented on drawbacks of the cranial nerve card game, including that it necessitated prior knowledge and mastery of the cranial nerves and their functions ( $N=27$ , 26%), the limited amount of time allotted for the game ( $N=24$ , 23%), confusion about the rules and how to

**TABLE 1** Students' evaluation of cranial chaos card game.

|                                                                                                   | N/A (%) | Strongly disagree (%) | Disagree (%) | Neither agree nor disagree (%) | Agree (%) | Strongly agree (%) | Median |
|---------------------------------------------------------------------------------------------------|---------|-----------------------|--------------|--------------------------------|-----------|--------------------|--------|
| The cranial nerve card game was easy to use                                                       | 4       | 4                     | 4            | 10                             | 81        | 85                 | 4.3    |
| The cranial nerve card game was interactive                                                       | 4       | 4                     | 1            | 10                             | 72        | 97                 | 4.4    |
| The cranial nerve card game facilitated a more practical understanding of cranial nerve functions | 4       | 8                     | 11           | 21                             | 59        | 85                 | 4.1    |
| Total                                                                                             | 12      | 16                    | 16           | 41                             | 212       | 267                | 4.3    |

**TABLE 2** Representative quotes for themes extracted from the open-ended comments from the survey using self-determination theory as a framework.

| Themes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>Competence</p> <ul style="list-style-type: none"> <li>• "This activity helped me to parse out which nerves were involved in/responsible for which motor and sensory functions."</li> <li>• "The card game was a major component of me feeling much more confident in my knowledge of cranial nerves when lab was finished."</li> <li>• "I definitely came away from it with a stronger understanding and foundation of the nerves."</li> <li>• "It was a fun way to strengthen my knowledge of the cranial nerves and had a positive long-term impact on my retention of the knowledge."</li> </ul>                                                                                                           |
| <p>Autonomy</p> <ul style="list-style-type: none"> <li>• "This game was fun and interactive, which was very helpful while trying to memorize a lot of information. I love it!"</li> <li>• "This was a very fun activity and a chance to actively utilize our knowledge in a fun and engaging way."</li> <li>• "The cranial nerve game allowed for us to apply what we learned in a fun setting. I genuinely think it made it easier and less overwhelming for me to remember the different functions of each cranial nerve."</li> <li>• "This was a fun way to interact with the material and made us think outside the box."</li> </ul>                                                                         |
| <p>Relatedness</p> <ul style="list-style-type: none"> <li>• "The interactive nature of this activity allowed me to speak with other students in my structure group which helped me learn some of the material."</li> <li>• "I enjoyed the collaborative aspect of this game with classmates."</li> <li>• "I didn't think much of the game at first, but it forced me to think of novel connections between the cranial nerves and our overall physiology."</li> <li>• "It helped me practice my knowledge by applying it to an activity that required thinking about the information in an unconventional way/different from the straight-forward information regurgitation I am used to practicing."</li> </ul> |
| <p>Drawbacks</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <p>Prior knowledge was necessary</p> <ul style="list-style-type: none"> <li>• "This activity assumed everyone memorized all the cranial nerves which was not the case at all."</li> <li>• "I was lucky to have studied the cranial nerves before so I found the game to be very helpful in applying that information, but I think other members on my team that had only done the prework the night before struggled to play the game since they had not really consolidated that information yet."</li> </ul>                                                                                                                                                                                                   |
| <p>Limited amount of time</p> <ul style="list-style-type: none"> <li>• "It was very quick, so I got a little bit out of it but if timing were better, we could've gotten through all of them."</li> <li>• "The idea was good, but the execution wasn't so great as we didn't really have enough time to go through an effective number of cards."</li> </ul>                                                                                                                                                                                                                                                                                                                                                     |
| <p>Confusion about the rules</p> <ul style="list-style-type: none"> <li>• "We jumped right into the card game without really understanding the rules/how to play. This made the first couple of rounds feel rushed and didn't particularly enhance my understanding of the cranial nerves."</li> <li>• "The only bummer about the game was the fact that there was no direct 'winner' to the specific questions because some answers could be 'argued' to be a better fit for the question. More specific rules/guidelines would be helpful."</li> </ul>                                                                                                                                                         |
| <p>Lack of variability of card options</p> <ul style="list-style-type: none"> <li>• "I think having a wider variety of 'action' cards would be beneficial."</li> <li>• "I think at times we repeatedly got the same cards or multiple of the same so mixing them up better might be helpful next time."</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                               |

choose the winner ( $N = 13$ , 13%), and the lack of variability of card options ( $N = 10$ , 10%).

## DISCUSSION

In this paper, we described the cranial nerve card game—an innovative and fun way to review cranial nerves. We also assessed students' perception of the game as a learning tool. Our findings were situated within the three components of the self-determination theory: competence, autonomy and relatedness.<sup>19</sup> The cranial nerve card game was highly rated as being easy to use, interactive, and improved their practical understanding of cranial nerve functions. Competence underscores the need to attain a sense of mastery, and we found that over 40% of students commented that the cranial nerve game aided in their knowledge of cranial nerve function. Students commented on how the game

challenged them to make connections between cranial nerves and their real-world function in unique ways, relating the structure to the function of the nerves. Relatedness was also expressed in terms of the team-based learning aspect of the session with students feeling connected with their peers. Finally, making learning fun can increase a sense of autonomy, by motivating students to be creative in the learning environment.

There have been reported decreases in student engagement with traditional teaching methods, resulting in rote memorization—the lowest process in Bloom's Taxonomy, bereft of intrinsic interest or application to relevant clinical settings.<sup>21</sup> Experimentation with various non-traditional pedagogical methods, including game-based learning, have been widely studied with positive outcomes,<sup>22–24</sup> hence the introduction of a card game as a non-traditional learning aid to strengthen the understanding of cranial nerve function through engagement of students in their own learning process.<sup>25</sup> The similarity in design of the cranial chaos to a conventional card

game made it a bit familiar and easy to use, with only about 7% expressing confusion about the rules of the game. To reduce confusion about the cranial chaos game rules, the rules will contain visual illustrations and be displayed on the projector throughout the gameplay.

While there are options for online and virtual-reality cranial nerve games,<sup>14,15</sup> none of them capture the functions of cranial nerves using real-life scenarios. Moreover, most would require expensive high-powered devices with high-speed internet connections and vast data storage capabilities. This can be a limitation to applying these tools in the classroom.<sup>26</sup> Besides, this would present a barrier to implementation by institutions with limited resources. Moreover, though digital games are excitable and immersive, the extra burden of navigating yet another technological system may distract from the educational focus of the game. Conversely, cranial chaos is a portable, analog, readily accessible cranial nerve card which can be easily adapted to various instructional settings while encouraging physical interactions and collaborative active learning.

The use of real-life scenarios in the game design provided an opportunity for the students to explore, reflect and apply their knowledge of cranial nerves in a meaningful way, while promoting critical thinking and collaborative team-based learning.<sup>27</sup> The randomness of the scenarios also drove the students to revise and validate their mental models of cranial nerves.<sup>28</sup> 10% of the students complained of limited variability in the scenarios, and so more scenarios will be added. The scenarios can also be tailored to reflect curricular priorities and include cultural contexts which make it easily adaptable for students across all health professions and international institutions.

Since this session was conducted as a flipped classroom, the students were provided with the learning objectives and suggested resources a week before so that they could prepare ahead of time, and so the card game served as a tool to solidify concepts and for practice. Thus, the cranial chaos game is not limited to use only as part of teaching but can be integrated into review sessions. However, 26% of the students either did not complete their pre-work or did not have a full grasp of the functions of the cranial nerves prior to the game. In response, the students proposed a brief overview of the functions of the 12 cranial nerves prior to the cranial chaos game. We intend to adopt this in future iterations of the session. In addition, the immediate feedback provided by the judges after each round was useful in closing some gaps in knowledge and further stimulating self-reflection, imperative for both learning and engaging.<sup>29,30</sup> Another major challenge encountered during the implementation of cranial chaos was timing, with 23% of the students complaining about limited time allocated to the game. Since the duration of each laboratory session was only 30min, about 15min was allocated to playing the game which allowed for only about three rounds, limiting the opportunity for exposure to more scenarios. To address this, we have now made the card decks available to students to use outside class, which would enable them to utilize it at their own pace for individual or group study and review.

Introduction of scores, and competition in serious games has been positively linked with increased extrinsic motivation in line with

the self-determination theory.<sup>31</sup> Understandably, 6% of the students did not like the fact that the game had no winner, as they did not find it challenging enough. Modifications, however, have been explored to establish a clear winner for each round based on the most accurate and creative responses, in order to create additional motivation while maintaining the fun element. Our initial concern was that a competitive element might mostly attract participation from extroverted and high-performing students which may negatively impact the participation of introverted and low-performing students, create a game of winners and losers and thus defeat the purpose of learning in a fun, low-pressure atmosphere.<sup>32</sup> Nevertheless, the incorporation of the bonus cards—Foramina Fumble, Synaptic Snag, and Pathway Peril was done in a way that mostly challenged the consolidation of other bony and pathway elements of cranial nerves. A fun environment fosters curiosity and experimentation, which promotes learning.<sup>28</sup> Fun as a critical component of game design was accomplished here as the students enjoyed playing and were quite invested in the experience. This was corroborated by the students' evaluation of the game and accompanying comments.

## CONCLUSION

Cranial chaos is a simple, readily accessible and adaptable teaching tool effective for reviewing cranial nerves using real-life scenarios. However, while self-reported measures such as the surveys used in this study offer useful insight into the learners' experience, they remain largely subjective and open to bias. Our future studies will therefore incorporate structured pre- and post- tests from both laboratory and clinical skills assessments for a more comprehensive and measurable evaluation of knowledge and clinical application of cranial nerves.

## AUTHOR CONTRIBUTIONS

**Shannon M. Knutson:** Conceptualization; data curation; investigation; methodology; writing – original draft; writing – review and editing. **Doreen M. Olvet:** Data curation; formal analysis; methodology; writing – original draft; writing – review and editing. **Vanessa M. Reddin:** Data curation; investigation; methodology; writing – review and editing. **Rosemary B. Bassey-Linus:** Data curation; investigation; methodology; project administration; writing – original draft; writing – review and editing.

## CONFLICT OF INTEREST STATEMENT

The authors have nothing to disclose.

## ETHICS STATEMENT

Ethical approval was obtained, and the study was approved under Exempt Review procedures of the Hofstra University Institutional Review Board (Ref #: 20220922-SOM-KNU-1).

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## SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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